



Modelling Financial Markets

Chapter 9: Realised Volatility



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BV: Bipower Variation
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ CLT: Central Limit Theorem
- ▣ DJIA: Dow Jones Industrial Average
- ▣ DM: Diebold–Mariano (test)
- ▣ ETF: Exchange-Traded Fund
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FOMC: Federal Open Market Committee
- ▣ FRED: Federal Reserve Economic Data
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR model with Continuous and Jump components
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ HEAVY: High-frEQUENCY-bAsed VolatilitY (models)
- ▣ IQ: Integrated Quarticity
- ▣ LLM: Large Language Model
- ▣ MAE: Mean Absolute Error
- ▣ MCS: Model Confidence Set

Acronyms used in this material (2/2)

- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ RF: Random Forest
- ▣ RK: Realised Kernel
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ SHAR: Semivariance HAR model
- ▣ TRV: Truncated Realised Variance
- ▣ TSRV: Two-Scale Realised Variance
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility IndeX
- ▣ VRP: Variance Risk Premium

Outline

Motivation

Realised Variance

Microstructure Noise

Jumps

Stylised Facts of Realised Volatility

The HAR-RV Model

Forecasting: HAR vs GARCH

Rough Volatility

Case Study: Machine Learning vs HAR

AI for Scientific Discovery

Summary

References

Volatility Is Latent: Can We Measure It?

The problem

- Volatility drives risk (Chapters 7–8), option prices (Chapter 12) and portfolio weights (Chapter 4)
- But it is never observed: one daily return is a single draw from a distribution whose dispersion we want to estimate
- GARCH (Chapter 5) **infers** today's variance from past daily returns through a model

The idea of this chapter

- Inside one day there are hundreds of price changes: add up their squares
- The sum **measures** the day's variance almost without a model: realised variance [Andersen & Bollerslev \(1998\)](#)
- Measured volatility can then be modelled and forecast like any time series
- Early idea: [Merton \(1980\)](#) showed that variance, unlike the mean, is estimated better with more frequent data

Realised Volatility: From Sceptics to Standard



Torben G. Andersen, 2012

Photo: Soralum (2012); CC BY-SA 4.0; Wikimedia

[Commons](#)

- Measure first, then model: Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001) study the distribution of realised volatility; Andersen, Bollerslev, Diebold & Labys (2003) forecast it directly
- Why the sceptics were wrong (Section 7) and textbook treatment: Franke, Härdle & Hafner (2019)

Data Used in This Chapter

Series	Frequency	Period	Days
SPY (ETF on the S&P 500)	5-minute bars, 09:30–16:00 New York time	2020–2026	1 489
Bitcoin	5-minute bars, 24-hour UTC days	2024–2026	659
SPY, Bitcoin	daily closes	2000/2014–2026	for GARCH
VIX	daily close	2020–2026	implied volatility

- Up to 18 September 2026; returns $r_i = 100(\ln P_i - \ln P_{i-1})$ in %; a bar is labelled by its start time
- SPY: 78 returns per full day, 42 on half-days; bars off the 5-minute grid removed; missing bars inside a day carry the last price
- Bitcoin: 288 returns per day; days with fewer than 95% of the bars dropped (88 of 747 calendar days, mostly a data gap in October 2025)

A Model for Intraday Prices

- Notation: day t has M intervals of length $\Delta = 1/M$, returns $r_{t,i}$, daily return $r_t = \sum_i r_{t,i}$
 - ▶ Variances in %²; annualised volatility $\sqrt{252 \times \text{variance}}$ (SPY), $\sqrt{365 \times \text{variance}}$ (Bitcoin); the SPY overnight return (close to next open) is outside the bars
- Log price in continuous time: $dp(s) = \mu(s) ds + \sigma(s) dW(s)$, with W a Brownian motion (Chapter 11)
 - ▶ $\sigma(s)$ is the **spot volatility**: it changes within the day and from day to day
- **Integrated variance** of day t : the quantity we want to measure
 - ▶ $IV_t = \int_{t-1}^t \sigma^2(s) ds$
- If σ is independent of W (no leverage) and the drift is zero (or fixed by the path of σ), then given that path the daily return is Normal: $r_t | \sigma \sim N(\int \mu, IV_t)$
 - ▶ With leverage (Section 6) this conditional normality fails, but the limit theory of RV below still holds
- IV_t is the natural daily variance; GARCH's σ_t^2 is a model-based forecast of it from yesterday's information

Definition: Realised Variance

- **Realised variance:** the sum of squared intraday returns

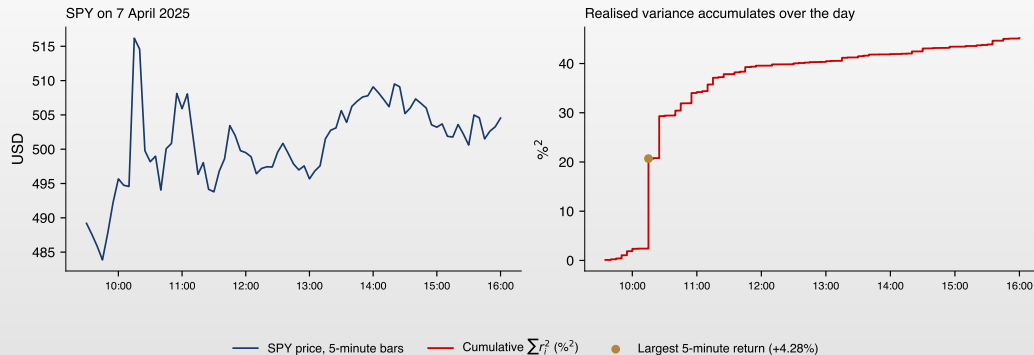
- ▶
$$RV_t = \sum_{i=1}^M r_{t,i}^2, \text{ realised volatility} = \sqrt{RV_t}$$

- As $M \rightarrow \infty$, $RV_t \rightarrow IV_t$: the drift vanishes and the sum of squares converges to the **quadratic variation** Barndorff-Nielsen & Shephard (2002)

- ▶ The mean of SPY 5-minute returns is below 1% of their standard deviation: it can be ignored

- No parameters to estimate: RV_t is a **model-free** measurement of one day's variance
- The squared daily return r_t^2 is the special case $M = 1$: unbiased (with zero drift) but extremely noisy

One Day, 78 Returns: 7 April 2025

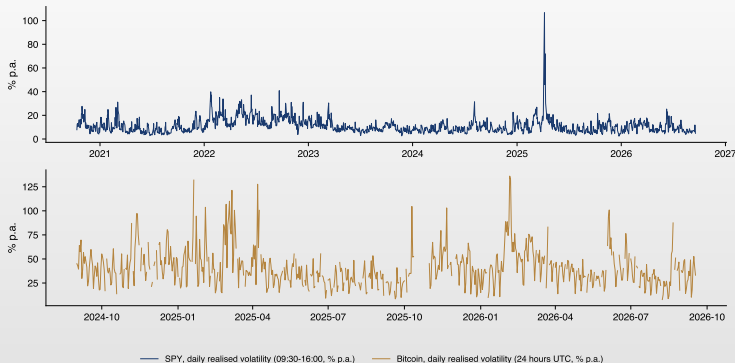


□ Left: SPY on 5-minute bars on the most volatile day of the sample; right: running sum of squared returns

Reading the Chart: 7 April 2025

- SPY opened -3.27% below the previous close, after tariff announcements; 78 five-minute returns followed
 - ▶ $RV = 45.1\%^2$: annualised realised volatility $\sqrt{252 \times 45.1} = 107\%$
- One 5-minute return of $+4.28\%$ (interval ending 10:15) contributed 41% of the day's realised variance
- Open-to-close the day ended $+3.09\%$: the daily return alone would call it a normal day, realised variance calls it one of the most turbulent days since 2020

Daily Realised Volatility: SPY and Bitcoin

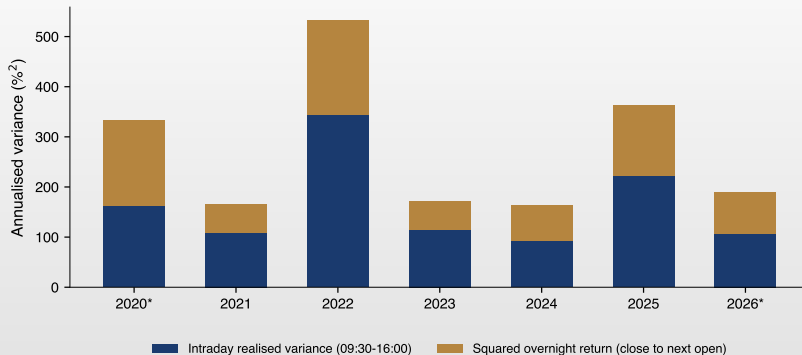


□ Annualised, from 5-minute returns; SPY: regular US session; Bitcoin: 24 hours; the gap in October 2025 is missing data

Reading the Realised Volatility Series

- SPY: volatility from mean RV $\sqrt{252 \overline{RV}} = 12.9\%$ p.a., median day 9.7%, range 2.4% to 106.7%
 - ▶ Volatility computed from mean variance exceeds the median: a few very volatile days raise the root-mean-square of daily volatilities
- Bitcoin: $\sqrt{365 \overline{RV}} = 45.1\%$ p.a., median 37.8%, maximum 136.1% on 2026-02-05
- Both series show **clustering**, as with GARCH (Chapter 5), but every point is a measurement, not a model output

Where Is the Risk: Day or Night?



- SPY, mean daily variance by year, annualised: intraday realised variance plus squared overnight return; 2020 and 2026 are partial years

Reading the Overnight Decomposition

- Overnight returns carry 38% of SPY's daily variance, although the market is closed for 17.5 hours
 - ▶ News (earnings, macro data, Asian and European sessions) accumulates and is priced at the open
- Annualised: close-to-close 16.6%; open-to-close 13.6%; overnight 10.1%
- Largest overnight move: -4.08% on 2024-08-05
- For daily risk we use **total** variance: $RV_t + r_{\text{overnight},t}^2$ (annualised mean 16.4%)
 - ▶ Alternatives: scale intraday RV up, or weight the two parts optimally [Hansen & Lunde \(2005\)](#)

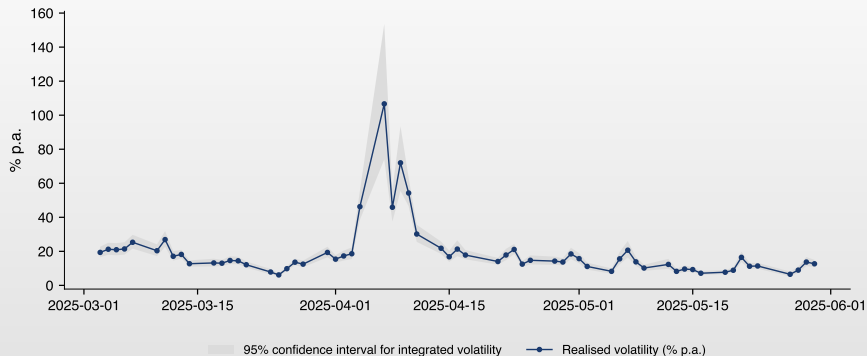
The Setting: Itô Semimartingales and Stable Convergence

- Itô semimartingale: $dp = \mu ds + \sigma dW + dJ$, with μ, σ adapted and σ càdlàg; σ may depend on W (leverage)
 - ▶ Without jumps: $\sqrt{M}(RV_t - IV_t) \xrightarrow{\mathcal{L}-s} MN(0, 2IQ_t)$, a **mixed** Normal distribution whose variance $2IQ_t$, with integrated quarticity IQ , $IQ_t = \int \sigma^4(s) ds$, is itself random [Barndorff-Nielsen & Shephard \(2002\)](#)
- **Stable** convergence ($\mathcal{L}$ -s): the limit holds jointly with every variable of the path's σ -field [Mykland & Zhang \(2009\)](#)
 - ▶ Hence $(RV_t - IV_t)/\sqrt{2RQ_t/M} \rightarrow N(0, 1)$: studentising by $\sqrt{2RQ_t/M}$, with $RQ_t \rightarrow IQ_t$ random, is legitimate only under stable convergence
- Finite samples: the t -statistic for $\ln RV_t$ has better coverage than the one in levels [Barndorff-Nielsen & Shephard \(2005\)](#); we use it on the next slide

How Precise Is Realised Variance?

- Feasible central limit theorem for RV [Barndorff-Nielsen & Shephard \(2002\)](#):
 $\sqrt{M}(\text{RV}_t - \text{IV}_t)/\sqrt{2\text{RQ}_t} \rightarrow N(0, 1)$
 - IQ_t is estimated by realised quarticity $\text{RQ}_t = \frac{M}{3} \sum_i r_{t,i}^4$
- Standard error: $\text{se}(\text{RV}_t) = \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$
- 95% CI on the log scale (always positive): $\text{RV}_t \exp(\pm 1.96 \text{se}(\text{RV}_t)/\text{RV}_t)$
 - With constant volatility, $\text{se}/\text{RV} \approx \sqrt{2/M}$: 16% for $M = 78$
- RV is a measurement **with error**: it matters when RV is the target of a forecast (Section 6)

Confidence Intervals for Daily Volatility

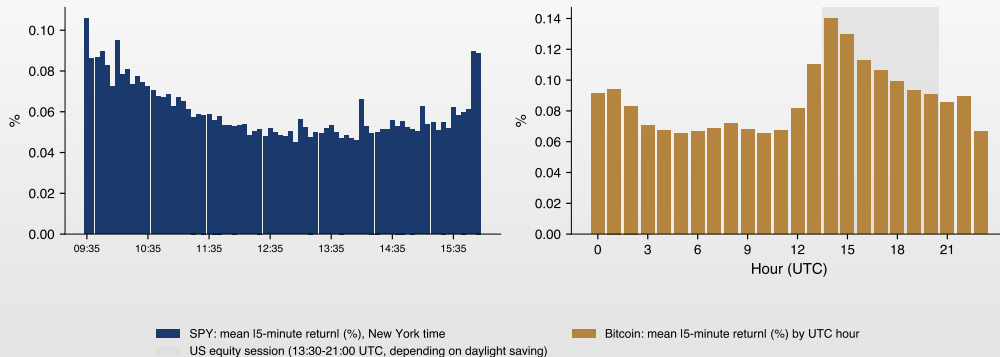


□ SPY, March–May 2025; band: 95% confidence interval for $\sqrt{252 IV_t}$ from realised quarticity

Reading the Confidence Intervals

- Median relative width of the interval (for volatility): 37% of its level
 - ▶ The band assumes a continuous price and negligible noise: on a jump day RV tends to $IV_t + \sum_j \kappa_j^2$, and $\sum r^4$ keeps the term κ^4 , so the band has no nominal coverage for either the total variation or IV_t (Section 4)
- 9 April 2025 (the day the tariff pause was announced): volatility 72%, CI [55%, 93%]
 - ▶ The interval widens when a few returns dominate the sum: $\sum r^4$ grows faster than $(\sum r^2)^2$
- Still far more precise than r_t^2 , whose relative error is of order $\sqrt{2} \approx 141\%$ under the Normal distribution
- Practical rule: a single day's volatility is known to about $\pm 20\%$ (95% band), averages over weeks much more precisely

Intraday Seasonality



Left: SPY, mean $|r_{t,i}|$ by 5-minute interval of the day (full days only); right: Bitcoin by UTC hour

Reading the Seasonality Pattern

- SPY: a **U-shape**: 0.106% in the first 5 minutes, 0.050% around midday, 0.089% in the last 5 minutes [Andersen & Bollerslev \(1997\)](#)
 - ▶ The open prices overnight news; the close concentrates index and fund rebalancing
- Bitcoin trades 24/7 but is most volatile at 14:00 UTC, near the US equity opening (09:30 New York: 13:30 UTC in summer, 14:30 UTC in winter); quietest at 10:00 UTC (ratio 2.2)
- RV sums over the day, so seasonality does not bias it; it does bias BV: $\mathbb{E}|r_i| |r_{i-1}| \propto \sigma_i \sigma_{i-1} \leq (\sigma_i^2 + \sigma_{i-1}^2)/2$, so fast changes of scale push RV – BV up and the jump test (Section 4) over-rejects
 - ▶ Remedy: divide $r_{t,i}$ by a periodicity factor f_i (standard-deviation estimator of [Boudt, Croux & Laurent \(2011\)](#)): $f = 1.78$ at the open, 0.73 at the minimum; SPY rejections at 5%: 235 $\rightarrow$ 167

Recap: Realised Variance

- ▣ $RV_t = \sum_i r_{t,i}^2$ measures integrated variance without a model; precision improves with M
- ▣ Realised quarticity gives its standard error: about $\sqrt{2/78} = 16\%$ of RV for one day of 5-minute data, i.e. a 95% band of roughly $\pm 20\%$ for volatility
- ▣ For SPY, overnight returns add 38% of daily variance: use total variance for daily risk
- ▣ Volatility follows a daily U-shape (SPY) and a UTC-hour pattern (Bitcoin)

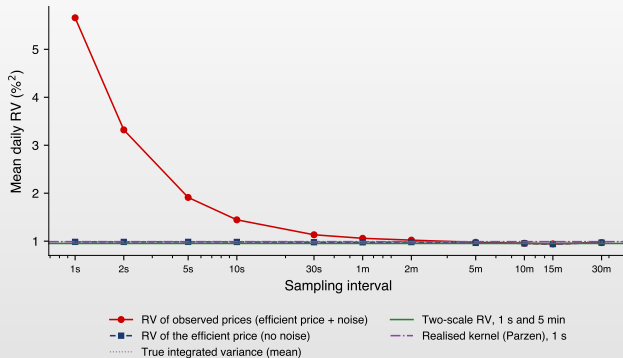
Why Not Use Every Tick?

- Observed prices are not the efficient price: $p_i^{\text{obs}} = p_i^* + \varepsilon_i$
 - ▶ Sources of ε : bid–ask bounce, tick size (price discreteness), stale quotes (Chapter 10)
- With i.i.d. noise of variance ω^2 and n returns per day [Zhang, Mykland & Aït-Sahalia \(2005\)](#), [Hansen & Lunde \(2006\)](#)
 - ▶ $\mathbb{E}[\text{RV}_t^{(n)} \mid \sigma] = \text{IV}_t + 2n\omega^2$ (zero drift, no leverage): the bias **grows** with the sampling frequency
 - ▶ Observed returns become negatively autocorrelated: $\text{cov}(r_i, r_{i-1}) = -\omega^2$
- Trade-off: more returns reduce sampling error but add noise bias

Deriving the Noise Bias and the Optimal Frequency

- Observed return $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$, with ε i.i.d. $(0, \omega^2)$ and independent of p^*
 - Given the path of σ (zero drift, no leverage: σ independent of W):
 $\mathbb{E}[\text{RV}^{(n)} \mid \sigma] = \sum_i \mathbb{E}[r_i^{*2} \mid \sigma] + \sum_i \mathbb{E}[(\varepsilon_i - \varepsilon_{i-1})^2] = \text{IV} + 2n\omega^2$; $\text{cov}(r_i, r_{i-1}) = -\mathbb{E}[\varepsilon_{i-1}^2] = -\omega^2$
 - By a law of large numbers for the noise terms, $\text{RV}^{(n)}/(2n) \xrightarrow{P} \omega^2$ as $n \rightarrow \infty$: the all-tick RV estimates the noise, not IV
[Bandi & Russell \(2008\)](#)
- MSE: $\mathbb{E}[(\text{RV}^{(n)} - \text{IV})^2] \approx \underbrace{2\text{IQ}/n}_{\text{discretisation}} + \underbrace{4n^2\omega^4}_{\text{squared bias}} + O(n)$
 - First-order condition $-2\text{IQ}/n^2 + 8n\omega^4 = 0$ gives $n^* = (\text{IQ}/(4\omega^4))^{1/3}$ [Bandi & Russell \(2008\)](#)
- Numbers: $\text{IV} = 1\%^2$, $\text{IQ} = 1$, $\omega = 0.01\%$: bias 0.016 at 5 minutes, 4.680 at 1 second; $n^* = 292$ (one return every 80 s)

The Volatility Signature Plot: Simulation



100 simulated days, one price per second; efficient price with random daily volatility plus i.i.d. noise, $\omega = 0.01\%$

Reading the Simulated Signature Plot

- True mean IV = 0.98; RV of observed prices: 5.66 at 1 second, 1.06 at 1 minute, 0.98 at 5 minutes
 - ▶ The bias matches $2n\omega^2$ from the previous slide
- Without noise, RV is flat in the sampling interval: the rise at high frequency is the fingerprint of noise
- Noise-robust estimators on 1-second data: two-scale RV 0.95, realised kernel 0.99
 - ▶ Error standard deviation across days: 5-minute RV 0.187, two-scale 0.150, kernel 0.045: using all data pays off only with a noise correction

Remedies: Sparse Sampling and Subsampling

- **Sparse sampling:** use one return every 5 minutes instead of every tick
 - ▶ Simple and robust; throws away most of the data [Liu, Patton & Sheppard \(2015\)](#)
- **Subsampling:** compute RV on K shifted grids (offsets $0, 1, \dots, K - 1$) and average
 - ▶ $RV_t^{\text{avg}} = \frac{1}{K} \sum_{k=0}^{K-1} RV_t^{(k)}$: same bias as one sparse grid, smaller variance
- **Two-scale RV (TSRV)** [Zhang, Mykland & Aït-Sahalia \(2005\)](#): remove the remaining bias with the all-data RV
 - ▶ $TSRV_t = RV_t^{\text{avg}} - \frac{\bar{n}}{n} RV_t^{\text{all}}$, $\bar{n} = (n - K + 1)/K$; divide by $1 - \bar{n}/n$ in small samples
 - ▶ $\bar{n}$ is the average number of returns per sparse grid; our grids also keep the open and the close, so $\bar{n}$ is their actual average count

Realised Kernels



Neil Shephard, 2004

Photo: Renate Schmid (2004); CC BY-SA
2.0 de; Wikimedia Commons

- **Realised kernel** Barndorff-Nielsen, Hansen, Lunde & Shephard (2008): add weighted autocovariances of returns
 - ▶ $RK_t = \gamma_0 + 2 \sum_{h=1}^H k\left(\frac{h}{H+1}\right) \gamma_h$, $\gamma_h = \sum_i r_{t,i} r_{t,i-h}$
 - ▶ Parzen weight function k : 1 at 0, falling smoothly to 0 at 1; keeps RK positive
- Noise makes γ_1 negative; the kernel adds it back and cancels the bias
- Bandwidth H grows with the noise-to-signal ratio: $H \approx 3.5 \xi^{4/5} n^{3/5}$, $\xi^2 = \omega^2 / \sqrt{IQ}$
- Shephard and co-authors developed kernels, bipower variation and the asymptotic theory of RV

Rates of Convergence under Noise

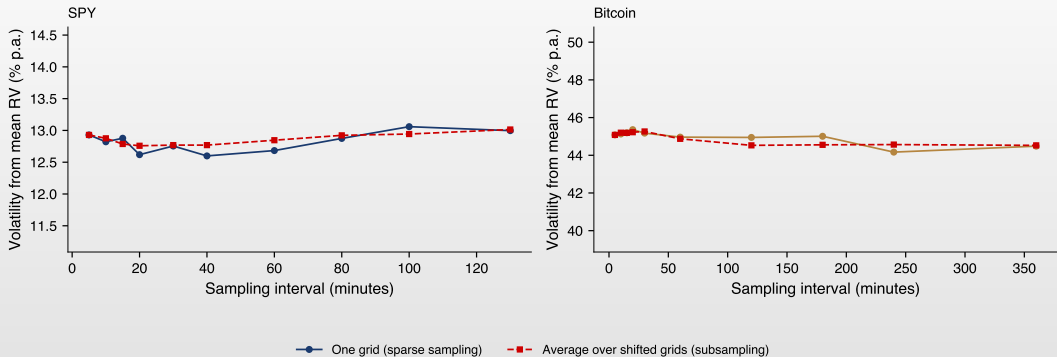
Estimator	Error of order	Positive	Source
RV on all ticks	inconsistent (bias $2n\omega^2$)	yes	Bandi & Russell (2008)
Sparse RV, fixed 5-minute grid	$M^{-1/2}$, M fixed; bias stays	yes	Liu, Patton & Sheppard (2015)
Two-scale RV	$n^{-1/6}$	no	Zhang, Mykland & Aït-Sahalia (2005)
Multi-scale RV	$n^{-1/4}$	no	Zhang (2006)
Realised kernel, flat-top / Parzen	$n^{-1/4}$ / $n^{-1/5}$	no / yes	Barndorff-Nielsen, Hansen, Lunde & Shephard (2008)
Pre-averaging	$n^{-1/4}$	no	Jacod, Li, Mykland, Podolskij & Vetter (2009)

- $n^{-1/4}$ is the optimal rate with i.i.d. noise: the price of noise is half the usual exponent
- With $n = 23\,400$ seconds: $n^{1/4} \approx 12.4$, against $\sqrt{78} \approx 8.8$ for 5-minute RV: a heuristic only (rates ignore constants, noise size and the bias of sparse RV), consistent with the modest gains in [Liu, Patton & Sheppard \(2015\)](#)

Pre-Averaging

- Local averages of $k_n \approx \theta\sqrt{n}$ returns with a weight g on $[0, 1]$, e.g. $g(x) = \min(x, 1 - x)$ Jacod, Li, Mykland, Podolskij & Vetter (2009)
 - ▶ $\bar{r}_i = \sum_{j=1}^{k_n-1} g(j/k_n) r_{i+j}$: noise variance shrinks like $1/k_n$, the signal keeps order k_n/n
- Estimator, with $\psi_1 = \int_0^1 g'(x)^2 dx$, $\psi_2 = \int_0^1 g(x)^2 dx$ (1 and 1/12 for the triangular g)
 - ▶ $\widehat{IV} = \frac{1}{\theta \psi_2 \sqrt{n}} \sum_i \bar{r}_i^2 - \frac{\psi_1}{2 \theta^2 \psi_2 n} \sum_i r_i^2$ (second term: noise bias correction)
- Rate $n^{-1/4}$; the same device gives jump-robust (pre-averaged bipower) and covariance estimators
- Built for tick data: on 5-minute bars little noise is left to average out (next slides)

The Signature Plot on Real Data



- Annualised volatility from mean RV by sampling interval: one grid vs the average over all shifted grids

Reading the Empirical Signature Plot

- SPY: 12.9% at 5 minutes, 12.8% at 30 minutes, 13.0% at 130 minutes: flat
 - ▶ First-order autocorrelation of 5-minute returns: -0.008 (SPY), 0.005 (Bitcoin): no visible noise at 5 minutes
- Bitcoin: 45.1% at 5 minutes, 44.5% at 6 hours: flat as well
- On 5-minute SPY data: two-scale $RV / RV = 0.97$, realised kernel / $RV = 0.95$; correlations with RV 0.89 and 0.85
 - ▶ The diagnostics show little noise bias at 5 minutes: the corrections move RV by a few per cent and add estimation variability
- Consistent with [Liu, Patton & Sheppard \(2015\)](#): almost 400 estimators on 31 assets rarely beat 5-minute RV; noise-robust estimators pay off when noise is large relative to the signal (tick or second-level data, illiquid assets)

Realised Covariance and Asynchronous Trading

- Realised covariance $\sum_i r_{t,i}^{(1)} r_{t,i}^{(2)} \rightarrow \int \sigma_{12}(s) ds$ needs prices of both assets on the **same** grid
 - ▶ Epps effect [Epps \(1979\)](#): measured correlation falls towards 0 as sampling gets finer, because trades are asynchronous and the last price is stale
- [Hayashi & Yoshida \(2005\)](#): sum the products of **all** pairs of returns whose time intervals overlap
 - ▶ $\sum_{i,j} r_i^{(1)} r_j^{(2)} 1\{(t_{i-1}, t_i] \cap (s_{j-1}, s_j] \neq \emptyset\}$: unbiased without synchronisation (in the absence of noise)
- With noise as well: pre-averaged or multivariate kernel versions; application: intraday portfolio risk (Chapter 4)

Recap: Microstructure Noise

- ▣ Noise adds $2n\omega^2$ to RV: sampling too often overstates volatility
- ▣ The signature plot is the diagnostic: RV should not depend on the sampling interval
- ▣ Remedies: 5-minute sampling, subsampling, two-scale RV ($n^{-1/6}$), flat-top realised kernels and pre-averaging ($n^{-1/4}$), Parzen kernel ($n^{-1/5}$)
- ▣ For liquid assets at 5 minutes (SPY, Bitcoin) plain RV is the benchmark

Prices Jump

- Add jumps to the model: $dp = \mu ds + \sigma dW + dJ$, with J a jump process (news, Chapter 11)
 - ▶ Quadratic variation now has two parts: $RV_t \rightarrow IV_t + \sum_j \kappa_j^2$, with κ_j the jump sizes in day t
- Why separate them? Jumps are less persistent than diffusive volatility, and they price differently (options, Chapter 12)
- We need an estimator of IV_t that ignores jumps

Bipower Variation



Ole E. Barndorff-Nielsen, 2007

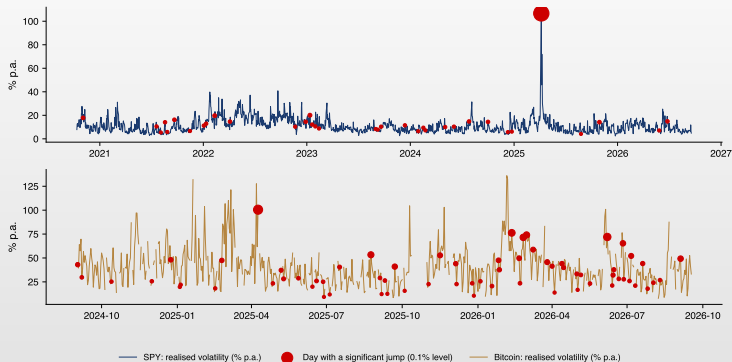
Photo: Thomas Steiner (2007); CC BY-SA 2.5;
Wikimedia Commons

- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns
 - ▶ $BV_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}| \rightarrow IV_t$
 - ▶ $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi} = 0.798$ for $Z \sim N(0, 1)$, so $\mu_1^{-2} = \pi/2 = 1.571$
- A jump enters one return: it is multiplied by a small neighbour, so its effect vanishes as M grows
- Jump part of the day: $\max(RV_t - BV_t, 0)$
- Barndorff-Nielsen (1935–2022) also shaped the Lévy-process models used for jumps

A Test for Jumps

- Ratio statistic [Huang & Tauchen \(2005\)](#), [Barndorff-Nielsen & Shephard \(2006\)](#):
$$z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\theta M^{-1} \max(1, TQ_t/BV_t^2)}}$$
 - ▶ $\theta = \pi^2/4 + \pi - 5 = 0.609$; tripower quarticity $TQ_t = M\mu_{4/3}^{-3} \frac{M}{M-2} \sum_i |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$, $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$
- Under no jumps $z_t \rightarrow N(0, 1)$; a jump makes $RV > BV$ and z_t large and positive
- One test per day: at level α we expect $\alpha \times$ days false alarms
 - ▶ We use $\alpha = 0.1\%$ (critical value 3.09): about 1.5 false jump days for SPY
- Significant day: jump part $J_t = RV_t - BV_t$, continuous part $C_t = RV_t - J_t$; otherwise $J_t = 0$

Jump Days: SPY and Bitcoin



□ Red points: days with a significant jump at the 0.1% level; point size proportional to the jump part J_t

Reading the Jump Test

- SPY: 35 jump days of 1 489 (2.4%); jumps are 2.8% of total realised variance
 - ▶ On a jump day the jump is on average 35% of that day's RV; BV/RV averages 0.95 over all days
 - ▶ Largest statistic: 2021-08-27 ($z = 5.2$), jump in the 10:00–10:05 interval, when the Federal Reserve chair's Jackson Hole speech was released
- Bitcoin: 67 jump days of 659 (10.2%), but only 1.8% of variance
 - ▶ Many small jumps on quiet days (weekends): one isolated move dominates a calm day
- The turbulent days of April 2025 are mostly **not** jump days: many large returns in a row raise BV as well

Many Days, Many Tests: Spurious Jumps

- 1 489 daily tests at 5%: 235 rejections against 74 expected under no jumps; which of them are false?
- [Benjamini & Hochberg \(1995\)](#): sort the p -values, reject the k smallest, with k the largest index such that $p_{(k)} \leq kq/N$; the FDR, $\mathbb{E}[\text{false/rejections}]$, stays $\leq q$ (with false/rejections $:= 0$ when nothing is rejected)
 - ▶ The guarantee needs valid p -values that are independent or positively dependent [Benjamini & Yekutieli \(2001\)](#); daily jump tests are only approximately so
 - ▶ SPY: FDR at 5%: 42 jump days; at 1%: 9; Bonferroni (FWER 5%): 7; FDR 5% after the periodicity correction: 17
- [Bajgrowicz, Scaillet & Treccani \(2016\)](#): without multiplicity control most detected jumps are spurious; the true ones are rare and cluster on news
- [Christensen, Oomen & Podolskij \(2014\)](#): at tick frequency most 5-minute “jumps” are steep but continuous bursts of volatility (fact or friction)

Locating Jumps Within the Day: Lee–Mykland

- Lee & Mykland (2008): $L_{t,i} = r_{t,i} / \hat{\sigma}_{t,i}$, with $\hat{\sigma}_{t,i}^2$ the bipower average of the previous $K = 270$ returns (their value for 5-minute data)
 - ▶ Under no jumps, $\max_i |L_i|$ over n returns has a Gumbel limit: reject if $(|L| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$, with $C_n = \sqrt{2 \ln n} - \frac{\ln \pi + \ln \ln n}{2\sqrt{2 \ln n}}$ and $S_n = 1 / \sqrt{2 \ln n}$ (our $\hat{\sigma}^2$ includes the factor $\pi/2$, so the constant $c = \sqrt{2/\pi}$ of Lee & Mykland (2008) becomes 1)
 - ▶ SPY, returns divided by f_i , $\alpha = 1\%$, $n = 115\,453$ (the whole sample): threshold $|L| > 5.41$
- 169 jumps on 105 days; 62 of them on Wednesdays between 14:00 and 15:00, when FOMC statements (14:00) and press conferences (14:30) are released
 - ▶ Largest: $L = -22.9$ on 2025-10-29; only 10% of these days are also flagged by the daily ratio test
- The two tests ask different questions: one extreme return relative to local volatility, versus a daily gap between RV and BV

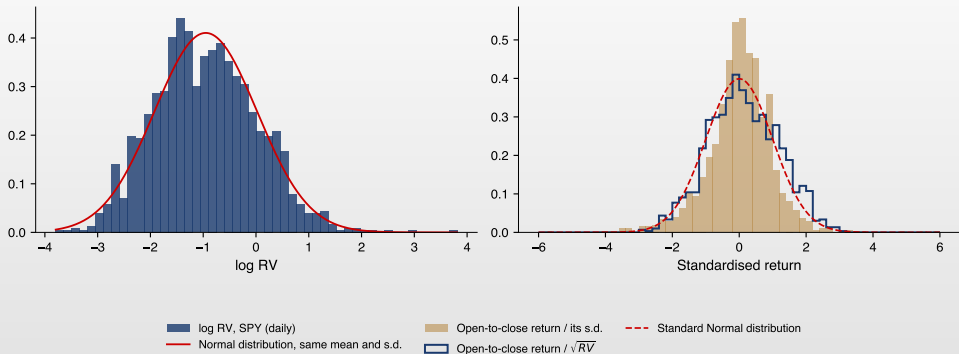
Jump-Robust Estimators of Integrated Variance

- Truncated RV [Mancini \(2009\)](#): $\text{TRV}_t = \sum_i r_{t,i}^2 \mathbf{1}\{|r_{t,i}| \leq c \hat{\sigma}_t f_i \Delta^\varpi\}$, $\varpi \in (0, 1/2)$
 - ▶ Diffusive increments are $O_p(\Delta^{1/2})$ and end up below the threshold; jumps are $O(1)$ and are cut off
- MedRV [Andersen, Dobrev & Schaumburg \(2012\)](#): $\frac{\pi}{6-4\sqrt{3}+\pi} \frac{M}{M-2} \sum_i \text{med}(|r_{i-1}|, |r_i|, |r_{i+1}|)^2$
 - ▶ An isolated return larger than both neighbours never enters the median (which shifts to a neighbour), so a large isolated jump is excluded, though its effect is not removed exactly in finite samples; MedRV is less sensitive than BV to isolated zero returns; two adjacent jumps can still enter the median
- Jump share of SPY RV: BV 6.3%, MedRV 7.2%, TRV 12.5% ($c = 3$, $\varpi = 0.49$; truncation bites on 44% of days: c is a tuning choice)
- [Aït-Sahalia & Jacod \(2009\)](#): compare power variations $\sum |r|^p$, $p > 2$, at Δ and $k\Delta$; the ratio tends to 1 with jumps and to $k^{p/2-1}$ without: known limits under both hypotheses

Recap: Jumps

- RV measures total quadratic variation: diffusion plus squared jumps
- Bipower variation is robust to jumps; $RV - BV$ estimates the jump part
- The ratio test flags about 2.4% of SPY days at the 0.1% level; jumps are a small share of total variance
- Test many days, control the FDR: 235 SPY days at 5% shrink to 42 with Benjamini–Hochberg; Lee–Mykland locates jumps in time (FOMC releases)
 - ▶ Jumps are separated from continuous variation in the HAR-CJ model (Section 5, Seminar 9)

Distributions: log RV and Standardised Returns

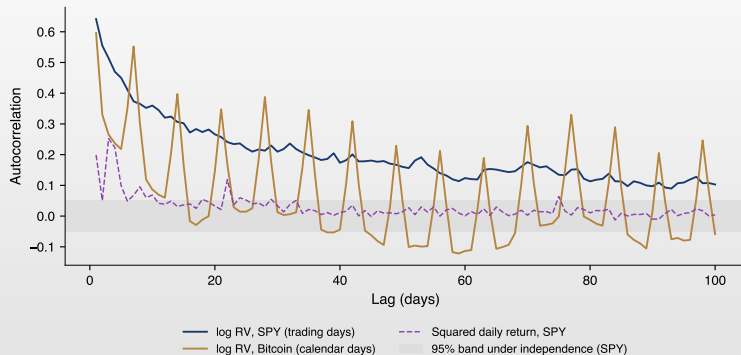


- Left: daily log RV of SPY and the Normal density; right: open-to-close returns scaled by their standard deviation (filled) and by $\sqrt{RV_t}$ (outline)

Reading the Distributions

- SPY RV: skewness 20.0, kurtosis 550; log RV: skewness 0.29, kurtosis 3.20 (the Normal distribution: 0 and 3)
 - ▶ Bitcoin log RV: skewness -0.27 , kurtosis 3.33: model volatility in logs
- Open-to-close returns: kurtosis 15.73; divided by $\sqrt{RV_t}$: 2.65, standard deviation 1.05
 - ▶ Bitcoin: from 7.57 to 2.77
- Heavy tails of daily returns (Chapter 1) come mostly, ex post, from changing volatility: approximately a **mixture** of Normal distributions with variances IV_t , exact only without leverage and jumps (Section 2)

Long Memory in Realised Volatility

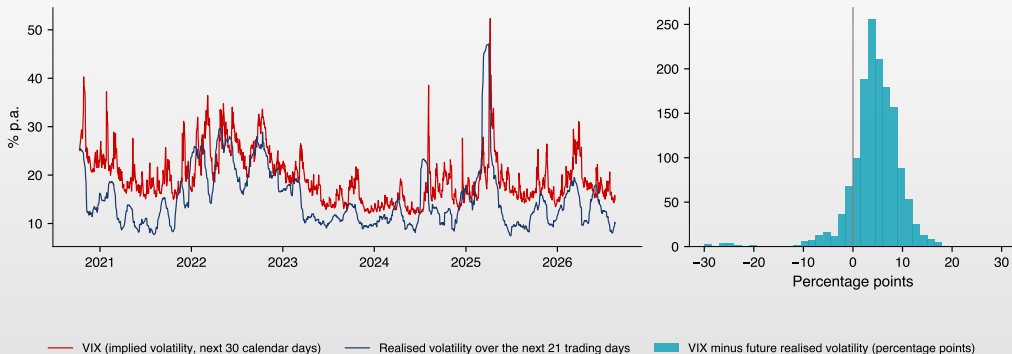


- Autocorrelations up to 100 days: log RV of SPY (total variance) and Bitcoin, squared daily SPY returns

Reading the Autocorrelations

- SPY log RV: 0.64 at lag 1, 0.24 at one month, still 0.10 at 100 days: **long memory**
 - ▶ An AR(1) would decay geometrically: $0.64^{22} \approx 0.00005$
- Squared daily returns show the same persistence, hidden by noise: 0.20 at lag 1
- Bitcoin: peaks at 7, 14, 21, ... days: a **weekly cycle**
 - ▶ Weekend RV is only 0.41 of weekday RV (Saturday 27% vs Wednesday 48% p.a.): no equity or macro news, fewer institutional traders

Implied vs Realised Volatility



- Left: VIX and SPY realised volatility over the next 21 trading days (including overnight returns); right: the gap between them

Reading the Variance Risk Premium

- VIX: the market's 30-day volatility, implied by S&P 500 options (Chapter 12); mean 19.5% vs 15.2% realised afterwards
 - ▶ VIX exceeded the subsequent realised volatility on 88% of 1 468 days; mean gap 4.4 percentage points
- **Variance risk premium (VRP)** Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009):
$$\text{VRP}_t = \mathbb{E}_t^Q[\text{RV}_{t,t+21}] - \mathbb{E}_t^P[\text{RV}_{t,t+21}]$$
, risk-neutral minus physical expectation of the same future variance; VIX^2 approximates the first term
 - ▶ Investors pay to insure against volatility spikes; the chart shows an **ex-post** spread (VIX minus the volatility that followed), a noisy proxy of this ex-ante premium
- As a forecast: regressing future variance on VIX^2 gives slope 0.56 (SE 0.05), $R^2 = 0.23$: VIX is informative but biased upwards
- The ex-post spread turns negative when realised volatility exceeds the earlier implied volatility (March–April 2025): a forecast error, not evidence of a negative ex-ante premium

Recap: Stylised Facts

- $\log RV$ is close to Normal Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001): model and forecast volatility in logs
- Returns divided by same-day $\sqrt{RV}$ are close to $N(0, 1)$ ex post: fat tails come mostly from changing volatility
- Long memory: the model must weigh the last month, not only yesterday
- Implied volatility carries a risk premium: it is an upward-biased forecast

The HAR-RV Model

- Corsi (2009): $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$
 - ▶ $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$ (week), $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$ (month)
 - ▶ Bitcoin trades every day: week = 7 days, month = 30 days
- Heterogeneous market hypothesis Müller et al. (1997): daily, weekly and monthly traders; a few components with different persistence mimic long memory
- An AR(22) with only 3 free slopes: weight $\beta_d + \beta_w/5 + \beta_m/22$ on lag 1, $\beta_w/5 + \beta_m/22$ on lags 2–5, $\beta_m/22$ on lags 6–22
- Estimation by OLS; errors are heteroskedastic and autocorrelated: use Newey–West (HAC) standard errors
 - ▶ log-HAR: the same regression for $\ln RV$; forecast $\exp(\hat{y} + \hat{s}^2/2)$, $\hat{s}^2$ the residual variance: the conditional mean if the errors are $N(0, s^2)$ (lognormal approximation)
 - ▶ For non-Normal errors use the smearing factor $\frac{1}{T} \sum_t e^{\hat{\varepsilon}_t}$ Duan (1983); with heteroskedastic errors both are approximations

Extensions of HAR

- ▣ **HAR-CJ Andersen, Bollerslev & Diebold (2007)**: separate continuous and jump parts as regressors
 - ▶ $\ln RV_{t+1} = \beta_0 + \beta_{Cd} \ln C_t + \beta_{Cw} \ln C_t^{(w)} + \beta_{Cm} \ln C_t^{(m)}$
 - ▶ $+ \beta_{Jd} \ln(1 + J_t) + \beta_{Jw} \ln(1 + J_t^{(w)}) + \beta_{Jm} \ln(1 + J_t^{(m)}) + \varepsilon_{t+1}$
 - ▶ $C_t^{(w)}, J_t^{(w)}$ ($C_t^{(m)}, J_t^{(m)}$): arithmetic means over the last 5 (22) days, as $RV_t^{(w)}$; logs are taken after averaging
- ▣ **HARQ Bollerslev, Patton & Quaedvlieg (2016)**: less weight on yesterday's RV when it is imprecise (high realised quarticity)
 - ▶ β_d becomes $\beta_d + \beta_Q \sqrt{RQ_t}$ with $\beta_Q < 0$
- ▣ Weekend or holiday dummies, leverage (negative returns raise future RV), implied volatility as an extra regressor
- ▣ Multi-step: regress $\sum_{j=1}^h RV_{t+j}$ directly on the same regressors

HAR Estimates: SPY and Bitcoin

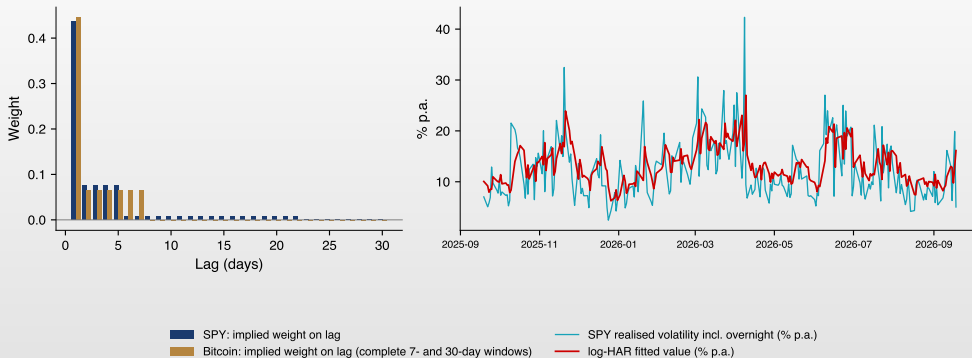
	β_0	β_d	β_w	β_m	R^2
SPY, HAR (levels)	0.273 (0.064)	0.399 (0.089)	0.222 (0.096)	0.114 (0.053)	0.31
SPY, log-HAR	-0.079 (0.027)	0.362 (0.037)	0.337 (0.048)	0.167 (0.044)	0.47
Bitcoin, HAR (levels)	1.496 (0.412)	0.412 (0.089)	0.409 (0.122)	-0.065 (0.097)	0.34
Bitcoin, log-HAR	0.223 (0.094)	0.380 (0.041)	0.465 (0.080)	-0.022 (0.098)	0.36

- Full samples; SPY: total daily variance (intraday RV plus squared overnight return), 1 467 days; Bitcoin: 24-hour RV, 547 days
- Newey–West standard errors in brackets

Reading the HAR Estimates

- SPY: all three components are significant; in logs $\hat{\beta}_d = 0.362$, $\hat{\beta}_w = 0.337$, $\hat{\beta}_m = 0.167$
 - ▶ Persistence $\hat{\beta}_d + \hat{\beta}_w + \hat{\beta}_m = 0.87$: shocks fade slowly, as in GARCH (Chapter 5)
- $R^2 = 0.47$ for log variance vs 0.31 for variance in levels: not comparable, the dependent variables differ; the out-of-sample losses on a common target (Section 7) decide
- Bitcoin: daily and weekly components matter; the monthly coefficient is close to 0 (-0.022 , SE 0.098)
 - ▶ Short sample (two years) and a strong weekly cycle: the week absorbs most of the memory

Implied Lag Weights and In-Sample Fit



- Left: weights of log-HAR on each past day (Bitcoin: complete 7- and 30-day windows); right: SPY, last 250 days, realised volatility and log-HAR fitted values

Reading the Lag Weights

- SPY: yesterday gets 0.437, each of days 2–5 gets 0.075, each of days 6–22 gets 0.008
- The staircase decays slowly: the month still holds weight $22 \times 0.008 \approx 0.167$
 - ▶ Three coefficients reproduce the long memory of the autocorrelation function
- Bitcoin: yesterday 0.446, days 2–7 about 0.066 each, almost nothing beyond a week
- In-sample fitted values follow the level of volatility, not the day-to-day noise

HAR as a Restricted AR(22): Testing the Restrictions

- Unrestricted: $\ln RV_{t+1} = \phi_0 + \sum_{k=1}^{22} \phi_k \ln RV_{t+1-k} + \varepsilon_{t+1}$
 - ▶ HAR imposes $\phi_2 = \dots = \phi_5$ (3 restrictions) and $\phi_6 = \dots = \phi_{22}$ (16): $R\phi = 0$ with 19 rows
- Wald: $W = (R\hat{\phi})^\top (R\hat{V}_{\text{NW}}R^\top)^{-1} R\hat{\phi} \rightarrow \chi^2_{19}$ under the HAR restrictions
- SPY log total variance: $W = 18.1$, $p = 0.51$; R^2 0.477 (AR(22)) vs 0.470 (HAR)
 - ▶ The staircase is not rejected: HAR saves 19 parameters at almost no in-sample cost, and less estimation noise helps out of sample

Measurement Error in the Regressors: Why HARQ

- $RV_t = IV_t + u_t$ with $\mathbb{E}[u_t | \sigma] = 0$ and $\text{Var}(u_t | \sigma) = 2IQ_t/M$ (the CLT of Section 2)
 - ▶ If $IV_{t+1} = \beta_0 + \beta IV_t + \eta_{t+1}$, OLS of RV_{t+1} on RV_t gives $\text{plim } \hat{\beta} = \beta \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}[2IQ/M]} < \beta$ for $\beta > 0$: attenuation
- If the signal variance does not rise with IQ_t (a working approximation), the signal-to-noise ratio of RV_t falls when IQ_t is high; HARQ approximates the falling weight by $\beta_d + \beta_Q \sqrt{RQ_t}$, $\beta_Q < 0$ [Bollerslev, Patton & Quaedvlieg \(2016\)](#)
- SPY, levels, Newey–West SE: intraday RV $\hat{\beta}_Q = -0.0088$ (0.0042); total variance -0.0022 (0.0024)
 - ▶ Daily slope (intraday RV) 1.13 at the median $\sqrt{RQ}$, 1.09 at its 99th percentile: the sign predicted by the theory, a small effect

Recap: HAR-RV

- HAR regresses tomorrow's RV on daily, weekly and monthly averages: OLS, three slopes
- It approximates long memory with a staircase of lag weights
- Estimate in logs; report Newey–West standard errors
- HAR is an AR(22) with 19 restrictions that the SPY data do not reject; measurement error in RV_t attenuates the daily slope, which HARQ corrects approximately
- Extensions: jumps (HAR-CJ), measurement error (HARQ), calendar effects, implied volatility

Forecasting Design

- Target: tomorrow's daily variance; SPY: total variance, 2022-01-03 to 18 September 2026 (1 180 days); Bitcoin: 2025-03-01 onwards (422 days)
 - ▶ Every forecast uses only data available the evening before
- Five models
 - ▶ log-HAR and HAR on realised variance, re-estimated daily on an expanding window
 - ▶ GARCH(1,1) with Student- t errors on daily returns since 2000 (SPY) or 2014 (Bitcoin), re-estimated every 21 days [Bollerslev \(1986\)](#)
 - ▶ EWMA with $\lambda = 0.94$ (Chapter 5) and "yesterday's RV" (random walk)
- HAR sees intraday data; GARCH and EWMA see only daily returns

Evaluating Against a Noisy Target

- True variance is unobserved; we compare forecasts F_t with the proxy RV_t
 - ▶ A loss is **robust** if the ranking by *expected* loss with a proxy that is conditionally unbiased (given the forecasts' information) equals the ranking with true variance [Patton \(2011\)](#)
- Two robust losses
 - ▶ $MSE = (RV_t - F_t)^2$: dominated by the most volatile days
 - ▶ $QLIKE = \frac{RV_t}{F_t} - \ln \frac{RV_t}{F_t} - 1 \geq 0$: depends only on the ratio; penalises under-prediction more
- Common mistake: MAE, or MSE on volatility instead of variance, is **not** robust
 - ▶ [Andersen & Bollerslev \(1998\)](#): against r_t^2 , GARCH forecasts explained a few % of the variation; against RV the same forecasts explain far more: the proxy was noisy, not the model

Tests: Diebold–Mariano and Mincer–Zarnowitz

- Diebold & Mariano (1995): loss difference $d_t = L(RV_t, F_t^A) - L(RV_t, F_t^B)$; $t = \bar{d}/\text{se}_{\text{NW}}(\bar{d}) \rightarrow N(0, 1)$
 - ▶ $t < 0$: model A has the lower loss; HAC standard error because d_t is autocorrelated (Chapter 8)
- Mincer & Zarnowitz (1969): $RV_t = a + b F_t + u_t$; an unbiased forecast has $a = 0$, $b = 1$
 - ▶ Joint Wald test with Newey–West covariance; R^2 measures how much of the variation the forecast explains
 - ▶ R^2 is capped by the noise in RV: even the true variance would not reach 1 Andersen & Bollerslev (1998)

Results: SPY, 2022–2026

Model	QLIKE	DM vs log-HAR (p)	MSE	MZ b	MZ R^2
log-HAR	0.310	–	4.15	1.16	0.30
HAR	0.326	–1.95 (0.052)	4.48	0.75	0.26
GARCH(1,1)- t	0.343	–2.30 (0.021)	4.87	0.70	0.20
EWMA	0.393	–4.02 (< 0.001)	5.14	0.74	0.14
Yesterday's RV	0.502	–5.95 (< 0.001)	5.41	0.54	0.29

□ 1180 one-day-ahead forecasts of total daily variance; DM: QLIKE differences, Newey–West; $t < 0$ favours log-HAR

Reading the SPY Results

- log-HAR has the lowest QLIKE and MSE: QLIKE 10% below GARCH(1,1)- t (DM $t = -2.30$, $p = 0.021$)
 - ▶ Against EWMA and yesterday's RV the gain is highly significant; against HAR in levels only marginally ($p = 0.052$)
- MSE differences are not significant: a few crisis days (April 2025) dominate the squared errors
- Mincer–Zarnowitz: log-HAR $b = 1.16$ ($p = 0.554$ for $a = 0$, $b = 1$), GARCH $b = 0.70$: no significant bias, but log-HAR explains more (R^2 0.30 vs 0.20)
- Yesterday's RV has a high R^2 but $b = 0.54$: right direction, far too extreme

Results: Bitcoin, 2025–2026

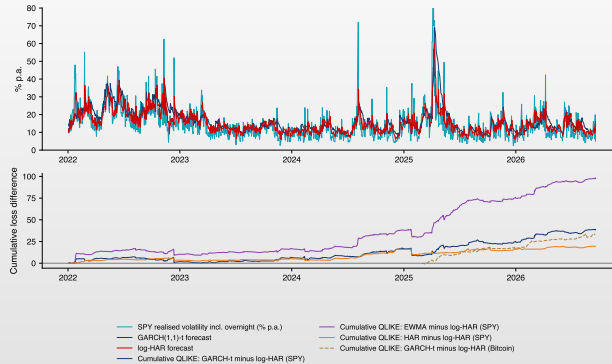
Model	QLIKE	DM vs log-HAR (p)	MSE	MZ b	MZ R^2
log-HAR	0.282	–	26.6	0.82	0.38
HAR	0.409	–1.22 (0.223)	27.6	0.89	0.34
GARCH(1,1)- t	0.360	–4.32 (< 0.001)	42.2	0.55	0.23
EWMA	0.346	–2.96 (0.003)	36.3	0.64	0.18
Yesterday's RV	0.466	–5.38 (< 0.001)	35.0	0.58	0.33

□ 422 one-day-ahead forecasts of 24-hour realised variance; same models and tests as for SPY

Reading the Bitcoin Results

- log-HAR wins again: QLIKE 22% below GARCH(1,1)- t (DM $t = -4.32$, $p < 0.001$)
 - ▶ GARCH on daily data is biased: MZ $b = 0.55$ ($p < 0.001$): it overreacts to large daily moves
- HAR in levels is worse than GARCH under QLIKE (0.409 vs 0.360): after calm weekends it predicts too low a variance, and QLIKE punishes under-prediction
- Intraday information can enter Bitcoin risk forecasts; entropy of high-frequency returns has been proposed as one such signal ([Pele & Mazurencu-Marinescu-Pele \(2019\)](#))
- Crypto-assets behave like a distinct asset class with their own volatility dynamics [Pele et al. \(2023\)](#)

Forecasts over Time

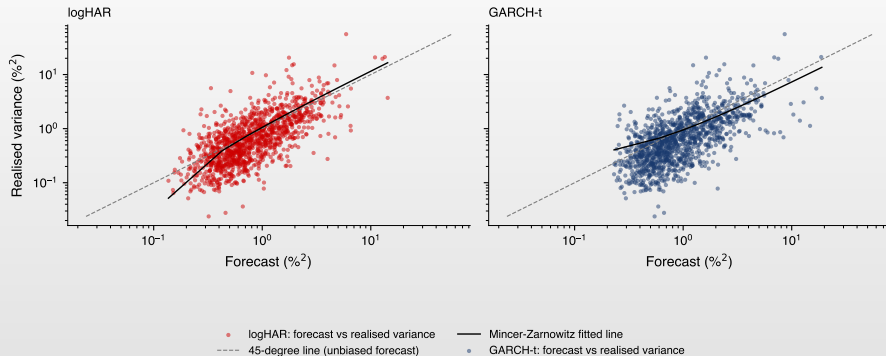


- Top: SPY realised volatility (with overnight) and the two forecasts, annualised; bottom: cumulative QLIKE loss of each model minus log-HAR (rising line = log-HAR better)

Reading the Forecast Paths

- After a shock both forecasts jump; GARCH stays high longer (its persistence is close to 1), log-HAR returns faster
 - ▶ April 2025: the GARCH forecast stayed above 40% p.a. until late April, while realised volatility was below 20% from mid-April
- The cumulative loss difference rises steadily, with jumps at turbulent episodes: the gain is not driven by one day
- Of the three models plotted against log-HAR, EWMA loses most: a fixed $\lambda = 0.94$ reacts too much to single days

Mincer–Zarnowitz Regressions



□ SPY, 2022–2026: realised variance against the forecast (log scales); dashed: 45-degree line; black: fitted regression in levels

Reading the Mincer–Zarnowitz Plots

- log-HAR forecasts span a wider range: they separate calm from turbulent days better
- Only 1% of GARCH forecasts are below $0.26\%^2$ per day (log-HAR: minimum $0.14\%^2$): on calm days realised variance falls far below GARCH
- The cloud is wide for both: one day's variance is hard to predict, and RV itself is measured with error
- Weekly horizon and model extensions: Seminar 9 (B6, B7)

Beyond HAR: Semivariances and Joint Return–RV Models

- Realised semivariances $RS_t^\pm = \sum_i r_{t,i}^2 \mathbf{1}\{r_{t,i} \gtrless 0\}$, $RS^+ + RS^- = RV$; SHAR [Patton & Sheppard \(2015\)](#): $\beta_d RV_t$ becomes $\beta^+ RS_t^+ + \beta^- RS_t^-$
 - ▶ SPY, levels (overnight return counted as one more return of the day): $\hat{\beta}^- = 0.56$ (0.23), $\hat{\beta}^+ = 0.21$ (0.18); Wald $\beta^+ = \beta^-$: $p = 0.35$
 - ▶ Bad volatility is more persistent, as in [Patton & Sheppard \(2015\)](#), but six years of data do not separate the two slopes
- Realized GARCH [Hansen, Huang & Shek \(2012\)](#): a GARCH equation driven by the realised measure x_t plus a **measurement equation**
 - ▶ $\ln h_t = \omega + \beta \ln h_{t-1} + \gamma \ln x_{t-1}$, $\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$, $r_t = \sqrt{h_t} z_t$: joint quasi-maximum likelihood
- HEAVY models [Shephard & Sheppard \(2010\)](#): one GARCH-type equation for the daily variance, driven by lagged RV, and one for RV itself: consistent multi-step forecasts

Which Models Survive? The Model Confidence Set

Model	QLIKE	MCS p -value	In MCS (90%)
log-HAR	0.310	1.000	✓
SHAR	0.320	0.224	✓
HAR	0.326	0.014	
HARQ	0.344	0.095	
GARCH(1,1)- t	0.343	0.095	
EWMA	0.393	0.012	
Yesterday's RV	0.502	0.000	

- Hansen, Lunde & Nason (2011): test equal predictive ability of all models (range statistic, block bootstrap), drop the worst, repeat; the surviving set contains the best model with asymptotic probability $\geq 90\%$ (under the method's regularity conditions)
- SPY, 1 180 days from 2022: the set is {log-HAR, SHAR}; HARQ and SHAR are estimated in levels with the filter of Bollerslev, Patton & Quaadvlieg (2016) (forecasts outside the in-sample range replaced by the mean)
- A multiple-comparison procedure: many pairwise DM tests without size control would overstate the evidence (Chapter 8)

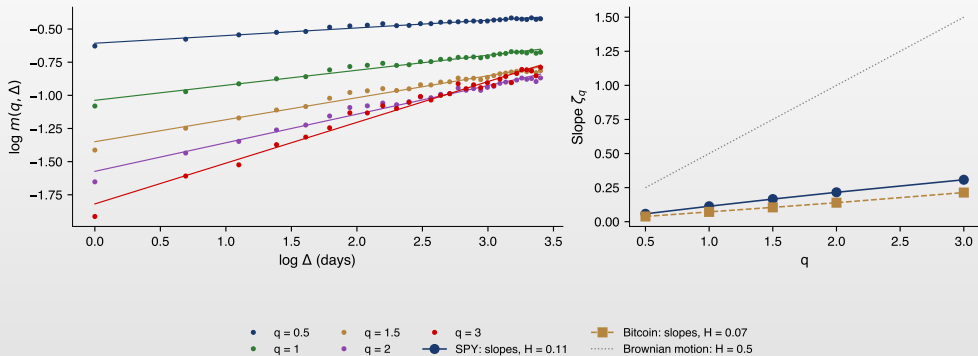
Recap: Forecasting

- Evaluate variance forecasts with robust losses (QLIKE, MSE on variance) and test differences with Diebold–Mariano
- Intraday data help: log-HAR beats GARCH(1,1)- t by 10% (SPY) and 22% (Bitcoin) in QLIKE
- Work in logs: HAR in levels loses under QLIKE when it under-predicts
- MSE rankings rest on a few crisis days and are rarely significant; the MCS keeps only log-HAR and SHAR for SPY

Is Volatility Rough?

- Gatheral, Jaisson & Rosenbaum (2018): log volatility behaves like a **fractional Brownian motion** with Hurst exponent $H \approx 0.1$
 - ▶ $H = 0.5$: ordinary Brownian motion; $H < 0.5$: increments are negatively correlated, paths are rougher; $H > 0.5$: smoother, trending
- Scaling law: $m(q, \Delta) = \frac{1}{T} \sum_t |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \propto \Delta^{qH}$
 - ▶ Regress $\ln m(q, \Delta)$ on $\ln \Delta$: slope $\zeta_q = qH$; then H from the slopes across q
- Rough models reproduce the shape of the implied volatility smile for short maturities (Chapter 12)

Scaling of Log Volatility Increments



□ Left: $\ln m(q, \Delta)$ against $\ln \Delta$ for SPY, $\Delta = 1, \dots, 30$ days; right: slopes ζ_q against q

Reading the Roughness Estimates

- Straight lines on the log–log scale: the scaling law holds; slopes are linear in q
 - ▶ $\hat{H} = 0.11$ for SPY (block bootstrap 95% CI [0.08, 0.13]), 0.07 for Bitcoin (only observed pairs at each calendar lag; interpolating the missing days gives 0.08): far below 0.5
- Caution: RV measures volatility with error, and measurement error lowers $\hat{H}$
 - ▶ A simulated Brownian motion gives $\hat{H} = 0.46$; plus noise of a similar relative size, $\hat{H} = 0.18$ (Seminar 9, B8)
- The scaling estimator on noisy RV proxies is **inconsistent** for the Hurst exponent; consistency needs high-frequency asymptotics [Fukasawa, Takabatake & Westphal \(2022\)](#)
 - ▶ Rough-looking estimates can come from estimation error in a non-rough model [Cont & Das \(2024\)](#); short-run roughness can coexist with long memory [Bennedsen, Lunde & Pakkanen \(2022\)](#)

Recap: Rough Volatility

- The moments of log-volatility increments scale like Δ^{qH} with $H \approx 0.1$
- Our data: $\hat{H} = 0.11$ (SPY) and 0.07 (Bitcoin)
- Part of the roughness can be measurement error in RV: i.i.d. error biases $\hat{H}$ down, other error structures can bias it either way, so $\hat{H}$ is not a bound on the latent H
- Rough volatility models are an active research area in option pricing

Case Study: Research Question

- Christensen, Siggaard & Veliyev (2023): does off-the-shelf machine learning (ML) forecast daily realised variance better than the HAR family, and if so, why?
 - ▶ HAR Corsi (2009) is linear in three regressors built from RV (daily, weekly and monthly averages), while RV is right-skewed and has long memory (stylised facts above): a nonlinear map may fit better
 - ▶ Its extensions (log-HAR, SHAR Patton & Sheppard (2015), HARQ Bollerslev, Patton & Quaedvlieg (2016)) still condition only on past returns
- Why other predictors rarely enter: OLS overfits when regressors are many, correlated and noisy
 - ▶ Candidates: implied volatility, the VIX, trading volume, macro and uncertainty indices (Table 1 of the paper)
- Why it matters: variance forecasts feed VaR (Chapter 7) and option pricing (Chapter 12); ML must beat a strong benchmark with three slope coefficients and an intercept
 - ▶ Design choice: minimal hyperparameter tuning, so the test is of the methods “as is”, not of the tuning effort

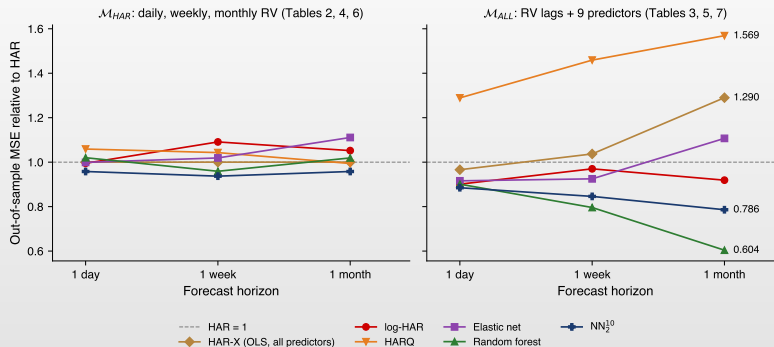
Case Study: Data and Method

- Data (Section 2): 29 of the 30 DJIA stocks (Visa excluded), 29 January 2001 – 31 December 2017, $T = 4\,257$ days; 5-minute RV from cleaned NYSE trade prices, $n = 78$ returns a day
 - ▶ Split: training 70% (2 964 days), validation 10% (424), test 20% (847); robustness: training 1 000 or 2 000 days, validation 200 (Appendix A.1)
 - ▶ $\mathcal{M}_{\text{HAR}}$: daily, weekly, monthly RV; $\mathcal{M}_{\text{ALL}}$ adds 9 predictors (Table 1): implied volatility, earnings-day dummy, 1-week momentum, change in log dollar volume; VIX, squared Hang Seng return, business-conditions index, change in the 3-month T-bill rate, policy-uncertainty index
- Models (Section 1): HAR, log-HAR, HAR with leverage, SHAR, HARQ by OLS on a rolling window; ridge (L_2 penalty), lasso (L_1), elastic net (a mix), adaptive and post-lasso, penalties tuned on the validation set
 - ▶ Trees: bagging, random forest (RF) with Breiman's defaults, 500 trees, $J/3$ candidate predictors per split (J : number of predictors), minimum node size 5 (Table A.6), gradient boosting
 - ▶ Neural networks NN_k^e : $k = 1, \dots, 4$ hidden layers (2; 4-2; 8-4-2; 16-8-4-2 neurons), leaky ReLU, Adam, dropout, early stopping; $e = 1$: best of 100 seeds on validation, $e = 10$: average of the best 10
- Evaluation (Section 1.6): out-of-sample MSE relative to HAR, averaged over stocks; one-sided DM test per stock [Diebold & Mariano \(1995\)](#); MCS at 90% [Hansen, Lunde & Nason \(2011\)](#)

Case Study: Key Results

- One day ahead, RV lags only (Table 2): MSE relative to HAR: log-HAR 0.995, HARQ 1.059, RF 1.020, neural networks 0.954–0.990
 - ▶ Past RV leaves little to learn: the networks give the clearest gains (1–5%), log-HAR barely beats HAR
- One day ahead, all 12 predictors (Table 3): OLS with all predictors 0.966, HARQ 1.289; elastic net 0.916, RF 0.901, log-HAR 0.901, NN_2^{10} 0.885
 - ▶ Extra predictors pay most with shrinkage, nonlinearity or the log transform; plain OLS gains less (0.966), and HARQ overfits
- One month ahead, all predictors (Table 7): RF 0.604, bagging 0.621, networks 0.786–0.964, log-HAR 0.919, OLS with all predictors 1.290
 - ▶ The gain grows with the horizon: the ML fits are more persistent (Figure 8) and approximate the long memory of RV better
- MCS at 90% (Figure 4): with all predictors, the networks stay in the set for about 80% of stocks, HAR for about 20–40%
 - ▶ Variable importance from accumulated local effects (Figure 7): daily RV, weekly RV and implied volatility dominate in all four models shown; RF and networks rank implied volatility higher than OLS with all predictors and elastic net

Case Study: Gains by Horizon

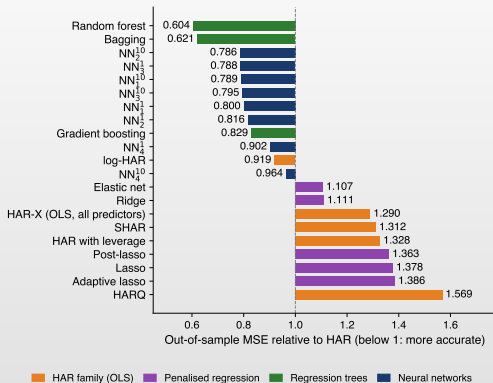


□ Source: Christensen, Siggaard & Veliyev (2023), Tables 2–7: out-of-sample MSE relative to HAR, averaged over 29 DJIA stocks (test set: the last 20% of 2001–2017); left: RV lags only, right: all 12 predictors

Case Study: Reading the Horizon Profile

- ▣ RV lags only: the six models shown stay between 0.937 and 1.111 times the HAR MSE at all three horizons
 - ▶ NN_2^{10} : 0.958, 0.937, 0.958; three RV averages leave little for a flexible model to learn
- ▣ All predictors: trees and networks improve with the horizon
 - ▶ Random forest $0.901 \rightarrow 0.796 \rightarrow 0.604$; NN_2^{10} $0.885 \rightarrow 0.846 \rightarrow 0.786$
 - ▶ The authors attribute this to the higher persistence of the ML fits, closer to the long memory of RV (Figure 8)
- ▣ The linear models given the same predictors deteriorate with the horizon
 - ▶ OLS with all predictors $0.966 \rightarrow 1.037 \rightarrow 1.290$, elastic net $0.916 \rightarrow 0.925 \rightarrow 1.107$, HARQ $1.289 \rightarrow 1.459 \rightarrow 1.569$
 - ▶ Only log-HAR stays below 1 (0.901, 0.970, 0.919): the log transform down-weights the high-variance days that dominate the MSE

Case Study: Ranking of 21 Models, One Month Ahead



Source: Christensen, Siggaard & Veliev (2023), Table 7: MSE relative to HAR, one month ahead, all 12 predictors, averaged over 29 DJIA stocks; colour: model family; NN_k^e: k hidden layers, e networks averaged

Case Study: Reading the Ranking

- The top two average many trees fitted on bootstrap samples: random forest 0.604, bagging 0.621; gradient boosting 0.829
 - ▶ All eight networks beat HAR (0.786–0.964), but depth does not help: the two four-layer networks are the weakest (0.902, 0.964)
- All five penalised regressions lose to HAR: ridge 1.111 and elastic net 1.107, the lasso variants 1.363–1.386
 - ▶ Linear HAR extensions with the extra predictors do worse still (from OLS with all predictors, 1.290, to HARQ, 1.569); log-HAR (0.919) is the only linear model below 1
- Each bar is an average of MSE ratios over stocks; Table 7 reports per-stock DM tests, not a test of the average ranking; the MCS of Figure 4 concerns one-day forecasts
 - ▶ Whether this ranking survives on SPY and under QLIKE is the question of Seminar 9, C3

Case Study: Replicate, Limits, Extend

- Replicate on course data: SPY 5-minute bars (data/market/intraday), open-to-close RV with 78 returns a day, as in Section 2; the authors report that SPY confirms the stock results (footnote 12)
 - ▶ Split of Appendix A.1: 1 000 training days, 200 validation days, the rest to 18 September 2026 as test set
 - ▶ Predictors: VIX (computed from S&P 500 index options; a proxy for SPY's own implied volatility, which would need SPY options), squared Hang Seng return, SPY momentum and dollar volume, T-bill and policy uncertainty (FRED); no earnings dummy for a fund (Seminar 9, C3)
- Limits stated by the authors
 - ▶ Minimal tuning: more tuning could improve ML further (Section 6); the networks are trained once and not re-estimated in the test window (Section 2)
 - ▶ Rankings depend on the split: with 1 000 training days the financial crisis falls in the test set and the HAR extensions beat HAR (Section 3.1, Appendix A.1)
 - ▶ HAR models with a separate jump component are not included (Section 1.2); accumulated local effects describe the average prediction and miss rare events such as earnings days (Section 3.2)
- Open extension: the paper ranks variance forecasts by MSE only; does the ML advantage survive QLIKE, which is robust to the noisy RV proxy [Patton \(2011\)](#), and does it hold for Bitcoin's 24-hour RV?

An Open Question: Is Volatility Really Rough?

- Gatheral, Jaisson & Rosenbaum (2018) report $H \approx 0.1$, but H is estimated from RV, a noisy proxy of the latent σ_t
 - ▶ Cont & Das (2024): the scaling estimator applied to realised volatility finds roughness even when spot volatility is smooth: an estimation artefact
 - ▶ Bolko, Christensen, Pakkanen & Veliyev (2023) (GMM that models the RV error) and Fukasawa, Takabatake & Westphal (2022) (consistent under high-frequency asymptotics) still find $H < 0.5$
- Formal question: $\ln \text{RV}_t^{1/2} = \ln \sigma_t + e_t$, with $\ln \sigma_t$ fractional of index H
 - ▶ H_0 : $H = 1/2$ (classical diffusive volatility) against H_1 : $H < 1/2$
 - ▶ Identification problem: a small H and a large $\text{Var}(e_t)$ both flatten $m(2, \Delta)$ at short lags
- Stakes: the short-maturity skew and volatility forecasts depend on H (Chapter 12)

The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: tabulate the evidence, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Tabulate the estimates of H published since 2018: estimator, sampling frequency, treatment of the RV measurement error, point estimate and standard error, DOI.
- 2. **Competing hypotheses**, each with its implied shape of $m(2, \Delta)$
 - ▶ Derive $m(2, \Delta)$ for three processes that give H near 0.1 with the scaling estimator: fractional log-volatility with $H = 0.1$; $H = 0.5$ plus i.i.d. error; a two-factor model with one fast mean-reverting factor.
- 3. **Data and code**: replicate the chapter's number first
 - ▶ Write Python that builds daily total variance from `data/market/intraday/SPY.US_5m.csv` (5-minute RV plus the squared overnight return, previous close to today's open), computes $m(q, \Delta)$ for q in $\{0.5, 1, 1.5, 2, 3\}$, $\Delta = 1..30$, and the slope H . Target: $\hat{H} = 0.11$

The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, pre-registered, with a power check
 - ▶ Write a pre-registered plan: lag windows [1,30], [5,30], [10,50]; RV, bipower variation and 30-minute subsampled RV; a fit $m(2, \Delta) = c \Delta^{2H} + k$; block-bootstrap intervals; size and power by simulation at $H = 0.1$ and 0.5 .
- 5. **Critique**: ask for the strongest counter-argument
 - ▶ Act as a referee: which non-i.i.d. error structures in log RV (serially correlated, level-dependent, driven by overnight returns or jumps) would survive the constant-k correction, and how would you detect each?
- 6. **Report**: all windows and measures, bootstrap intervals, the simulation design, code and the AI log

What the Researcher Must Check

- **Proxy error:** RV is an estimate with a known standard error; test: simulate $H = 0.5$ plus error of the same size and check that the procedure does not return $H \approx 0.1$
- **Overlapping increments:** $m(q, \Delta)$ reuses the same days at every Δ , so OLS standard errors of ζ_q are invalid; test: block bootstrap
- **Microstructure noise and jumps:** check the signature plot, then repeat with BV and with subsampled RV
- **Specification search:** lag windows and q grids chosen after seeing $\hat{H}$ inflate significance; fix them in advance and report all
- **Invented estimates:** assistants quote " $H \approx 0.1$ for all assets"; check each number against the table of the cited paper

Mini-Case: Is $\hat{H}$ an Artefact of i.i.d. Error?

- A typical assistant proposal: $\hat{H} \approx 0.1$ comes from i.i.d. measurement error in RV. Implication: $m(2, \Delta) = c \Delta^{2H} + k$ with $k = 2 \text{Var}(e_t) > 0$, so $\hat{H}$ rises when short lags are dropped

	$\Delta \in [1, 30]$	[5, 30]	[10, 50]	Fit with k , [1, 50]
Simulated $H = 0.5$ plus error	0.18	0.24	0.26	0.39
SPY, total RV	0.106	0.089	0.077	0.096
SPY, intraday RV	0.129	0.094	0.078	0.113

- First column replicates the chapter; the simulated row (one path, not a power study) shows the mechanism: $\hat{H}$ rises and the fit recovers most of the gap to 0.5
- SPY: $\hat{H}$ falls with the window and the fitted k is 0; $\hat{H} = 0.096$, block bootstrap 95% CI [0.058, 0.268]
- Interpretation: an illustrative diagnostic, inconsistent with the i.i.d.-error explanation; a formal rejection needs a simulation-calibrated test; the non-i.i.d. mechanism of [Cont & Das \(2024\)](#) is not tested by this design. Code: `Quantlets/Ch_09/ai_discovery_case.py`

Your Turn: A Project Seed

- Question: does the roughness of SPY and Bitcoin volatility survive an estimator that models the RV measurement error?
- Replicate first: $\hat{H} = 0.11$ (SPY) and 0.07 (Bitcoin) from the scaling law, and $\hat{H} = 0.18$ for simulated $H = 0.5$ plus error (Seminar 9, B8)
- Deliverables
 - ▶ A pre-registration (hypotheses, estimators, lag windows, tests, simulation design) dated before the first run
 - ▶ The GMM estimator of [Bolko, Christensen, Pakkanen & Veliyev \(2023\)](#) on the course RV series, with Monte Carlo size and power at $H \in \{0.1, 0.5\}$
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in AI_USE.md and every error the assistant made in AI_ERRORS.md: github.com/danpele/MFM/tree/main/project

Summary

- ▣ Realised variance turns intraday prices into a measurement of daily variance, with a known standard error
- ▣ Sample at 5 minutes for liquid assets; use noise-robust estimators when noise is material (tick data, illiquid assets); add the squared overnight return for daily risk
- ▣ Bipower variation, MedRV and truncation separate jumps from continuous variation; with FDR control jumps are rare and a small share of variance
- ▣ log RV is nearly Normal and has long memory: the log-HAR model captures it with three coefficients
- ▣ Out of sample, log-HAR beats GARCH for SPY and Bitcoin under QLIKE; MSE comparisons are inconclusive

Which Tool When?

Situation	Suggested tool
Liquid asset, 1–5 minute bars	RV at 5 minutes, subsampled
Tick data, or an illiquid stock (BVB)	Pre-averaging or flat-top realised kernel ($n^{-1/4}$; Parzen $n^{-1/5}$); check the signature plot
Daily risk of an exchange-traded asset	Total variance: RV plus squared overnight return
Separate news jumps from diffusion	BV or MedRV, ratio test with FDR control; Lee–Mykland for the jump time
Daily or weekly variance forecast	log-HAR (plus jumps, weekends, RQ); benchmark GARCH
Comparing forecasts	QLIKE and MSE on variance, Diebold–Mariano, Mincer–Zarnowitz, model confidence set

Coming Next: Market Microstructure (Chapter 10)

- ▣ Where does the noise in high-frequency prices come from? The limit order book, bid–ask spreads, tick sizes
- ▣ Liquidity and market impact: how prices react to trading
- ▣ High-frequency trading and crypto exchange order books
- ▣ Further reading: Andersen, Bollerslev, Diebold & Labys (2003), Corsi (2009), Franke, Härdle & Hafner (2019)

Further Study on Quantinar

Advanced courses on quantinar.com

-  Nonparametric and Semiparametric Models
-  Kalman Filter
-  Applied Time Series Analysis with Python

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

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